

Topic 4: Materials

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic.

Mathematical skills that could be developed in this topic include determining the slope of a linear graph and calculating or estimating, by graphical methods as appropriate, the area between a curve and the x-axis and realising the physical significance of the area that has been determined.

This topic may be studied using applications that relate to materials, for example spare-part surgery.

Students should:	
49.	be able to use the equation density $\rho = \frac{m}{V}$
50.	understand how to use the relationship upthrust = weight of fluid displaced
51.	a. be able to use the equation for viscous drag (Stokes' Law), $F = 6\pi\eta r v$. b. understand that this equation applies only to small spherical objects moving at low speeds with <i>laminar flow</i> (or in the absence of <i>turbulent flow</i>) and that viscosity is temperature dependent
52.	CORE PRACTICAL 4: Use a falling-ball method to determine the viscosity of a liquid.
53.	be able to use the Hooke's law equation, $\Delta F = k\Delta x$, where k is the stiffness of the object
54.	understand how to use the relationships (<i>tensile or compressive</i>) stress = force/cross-sectional area (<i>tensile or compressive</i>) strain = change in length/original length - Young modulus = stress/strain
55.	a. be able to draw and interpret force-extension and force-compression graphs b. understand the terms <i>limit of proportionality</i> , <i>elastic limit</i> , <i>yield point</i> , <i>elastic deformation</i> and <i>plastic deformation</i> and be able to apply them to these graphs
56.	be able to draw and interpret tensile or compressive stress-strain graphs, and understand the term <i>breaking stress</i>
57.	CORE PRACTICAL 5: Determine the Young modulus of a material
58.	be able to calculate the elastic strain energy E_{el} in a deformed material sample, using the equation $\Delta E_{el} = \frac{1}{2} F\Delta x$, and from the area under the force-extension graph <i>The estimation of area and hence energy change for both linear and non-linear force-extension graphs is expected.</i>